Continuous Nonabelian Collar Charging

An exact geometric charge and an ultraviolet compatibility no-go

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Abstract

We give a deterministic, continuous SU(2) counterpart to center-contour charging. For plaquettes $P_i = US_i$ incident on a selected link, relative staple distances equal relative plaquette distances for every value of the selected link. Failure of a common staple ball of radius η therefore forces an incident curvature above $\eta/2$ and total incident squared curvature above $\eta^2/2$. Combining this exact charge with the coherent- collar requirement $\eta < \rho/\sqrt{m}$ proves that the strongest local charge at the frozen ultraviolet witness has exponent only 0.9339, versus the required rate near seven. Thus the simplest one-link charged-collar repair is quantitatively incompatible with the declared parameters.

1 Relative geometry

Orient the m plaquettes incident on a link so that

$$P_i = US_i,$$

where U is the selected-link variable and S_i is its staple. Let d be the bi-invariant geodesic distance on SU(2).

Theorem 1.1 (Link-uniform collar charge)**.** *If the staple family is not contained in any common geodesic ball of radius η, then for every $U \in \mathrm{SU}(2)$,*

$$\max_i d(I, P_i) > \frac{\eta}{2}, \tag{1}$$

$$\sum_{i=1}^{m} d(I, P_i)^2 > \frac{\eta^2}{2}. \tag{2}$$

Both statements are gauge covariant.

Proof. Bi-invariance gives

$$d(P_i, P_j) = d(US_i, US_j) = d(S_i, S_j).$$

If the staple diameter were at most η, choosing any staple as a center would put the family inside an η-ball. Collar failure therefore supplies i, j with $D = d(P_i, P_j) > \eta$. Set $a = d(I, P_i)$ and $b = d(I, P_j)$. The triangle inequality gives $a + b \geq D$, so $\max(a, b) > \eta/2$. Moreover

$$a^2 + b^2 \geq \frac{(a+b)^2}{2} > \frac{\eta^2}{2},$$

which proves (2). Gauge covariance follows because all distances are bi-invariant. □

The theorem is stronger than a center-sector statement: no discreteness, commutativity, or coordinate chart is used.

2 Compatibility with the good collar

The coherent conditional theorem requires

$$\eta < \frac{\rho}{\sqrt{m}}. \tag{3}$$

At the limiting admissible radius, (1) charges only curvature

$$r_{\max} = \frac{\rho}{2\sqrt{m}},$$

and (2), inserted into a heat-kernel exponential $e^{-E/(2t)}$, supplies exponent at most

$$K_{\text{energy,max}} = \frac{\rho^2}{4mt}. \tag{4}$$

For a three-dimensional isotropic Gaussian radial event, put

$$J_3(s) = \begin{cases} \frac{1}{2}(s - 3 - 3\log(s/3)), & s > 3, \\ 0, & s \le 3. \end{cases}$$

The single-charge scaled radius is $s_{\max} = \rho^2/(4mt)$.

Corollary 2.1 (Local-charging incompatibility)**.** *At $t = 3 \times 10^{-6}$, $\rho = 0.0082$, and $m = 6$,*

$$s_{\max} = K_{\text{energy,max}} = 0.933888\ldots, \qquad J_3(s_{\max}) = 0.$$

Consequently local collar charging cannot supply either the declared $K = 7$ or the exact Stage 14 minimum $K = 6.665348\ldots$.

For a total-energy target K, compatibility requires

$$t \le \frac{\rho^2}{4mK}. \tag{5}$$

At $K = 7$ this is $t \le 4.00238 \times 10^{-7}$. Solving $J_3(s) = 7$ gives $s = 23.12719\ldots$ and the stricter radial requirement $t \le 1.21142 \times 10^{-7}$.

3 Interpretation and frontier

The result does not say collar failure is unchargeable. It proves the exact opposite and then shows that this local charge is too weak at the chosen scale. The charge must become extensive through multi-link integration, a coarser collar, a typed center/topological contour, or a smaller-time witness.

No probability estimate for the complete interacting measure is inferred from deterministic geometry alone. No RG iteration, continuum limit, or mass gap follows. The result is a C2 restricted theorem and no-go calibration.